

- 2 (a) State the **two** conditions necessary for a body to be in equilibrium.

1.	The net force is zero. [B1]
2.	The net torque is zero./ The net moment about any point is zero. [B1]

[2]

- (b) Fig. 2.1 shows a uniform beam AB of length 6.0 m and weight 2700 N suspended by two ropes AC and BC, each of length 6.0 m. The tensions in ropes AC and BC are T_1 and T_2 respectively.

A worker of weight 900 N is holding onto the beam at point D, where AD = 4.0 m and DB = 2.0 m.

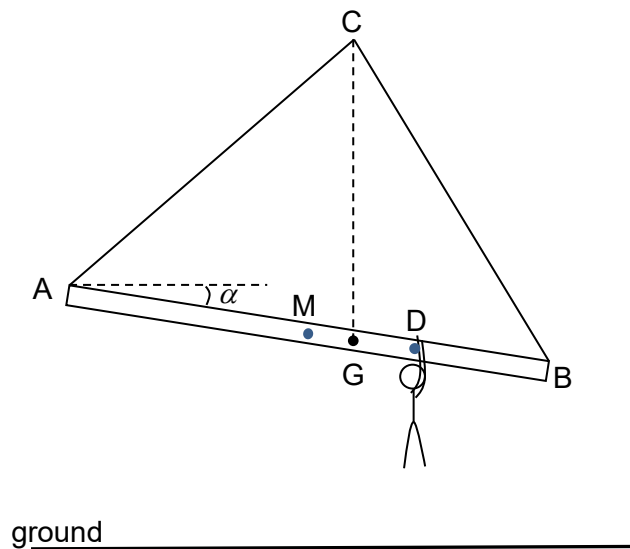


Fig. 2.1

The beam makes an angle α to the horizontal. The point M is the mid-point of the beam and the point G on the beam is the position of the centre of gravity of the beam and the worker.

- (i) Explain in terms of forces acting on the beam, why the point G must lie directly below C.

The combined weight of the beam and the worker can be considered to be acting at G. Considering moments about C, moments due to T_1 and T_2 about C is zero as their lines of action pass through C. Hence for the resultant moments about C to be zero, the line of action of the combined weight of the beam and worker must pass through C as well so that the moment of the combined weight about C is zero. Hence the vertical line through G must pass through C.

[2]

(ii) Calculate the distances MG and DG.

Considering moments due to the weight of beam, W_b , and that of the worker, W_w , about C,

Let distance MG be x and distance DG be y

$$W_b \cdot x = W_w \cdot y$$

$$(2700) x = (900) y$$

$$y = 3 x$$

$$MD = AD - AM = 4.0 - 3.0 = 1.0 \text{ m}$$

$$MD = MG + GD = x + 3 x = 1.0 \text{ m}, 4x = 1.0 \text{ m}, x = 0.25 \text{ m.}$$

$$MG = 0.25 \text{ m}$$

$$DG = 0.75 \text{ m}$$

distance MG = m

distance DG = m
[2]

(iii) If the angle α is 2.8° , determine the magnitude of the tension T_2 .

For horizontal equilibrium,

$$T_1 \cos (60.0^\circ - 2.8^\circ) = T_2 \cos (60.0^\circ + 2.8^\circ)$$

$$T_1 = \left(\frac{\cos 62.8^\circ}{\cos 57.2^\circ} \right) T_2$$

$$T_1 = 0.8438 T_2 \quad \dots (1)$$

For vertical equilibrium,

$$T_1 \sin 57.2^\circ + T_2 \sin 62.8^\circ = 2700 + 900 \quad \dots (2)$$

Substituting equation (1) into equation (2) and solving for T_2 ,

$$T_2 = 2250 \text{ N (3 s.f.)}$$

tension $T_2 = \dots \text{ N [2]}$

[Total: 8]

